

Paper 3.2: Persistent Native Memory for RAG

Transport, Composition, and Cross-Document Contextualization

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September 2026

Abstract

Retrieval-augmented generation repeatedly serializes selected evidence into a prompt and prefills it through the model. Progressive Retrieval Attention (PRA) instead retains selected evidence as reusable layer-wise attention K/V. We freeze retrieval output to separate evidence selection from native realization, positional transport, and cross-document composition. Across five natural MultiHop-RAG cohorts, all 600 matched contiguous text/native executions preserve generated output, first-step logits, and answer NLL. Warm native reuse reduces TTFT by 92.3% and total latency by 81.5% relative to re-prefilling the same selected text. An exact ordinary prefix-cache hit remains faster, but when the selected set changes, exact-prefix reuse falls to zero while PRA reuses 28.1% of selected K/V.

A selector-frozen decomposition over 150 questions then localizes the failure of arbitrary record composition. Independently encoded pre-RoPE records rebound to packed positions closely track a packed control in which cross-document attention is disabled (.108 versus .110 F1; first-step JS 8.80×10^{-4}). Shape-matched host-RoPE rebinding is exact across FP32, FP16, INT8, and INT4. Position can therefore be transported, but contextual state that was never formed cannot be recovered by rotation alone. Fixed partial materialization, parameter-free gist and boundary repair, and fixed sparse interaction do not close this gap. A 1.38M-parameter rank-8 residual raises held-out independent-PRA F1 from .135 to .154, but its five-seed interval crosses zero and answer NLL worsens. Persistent contiguous transport is established in the measured setting; cheap independent-record composition remains open.

1 Introduction

RAG systems search an external corpus, select a bounded evidence set, serialize that set as text, and prefill the model. Retrieval must expose the right evidence, and the model must repeatedly process that evidence when resources recur across requests. Ordinary prefix caches help only when the serialized prefix is identical.

PRA moves this reuse boundary. A selected resource can be retained as model-native K/V and addressed later without appearing again as visible prompt text. Here, *native memory* means reusable layer-wise attention K/V state, not retrieved text that must be re-prefilled. This is attractive for changing queries and document combinations, but it exposes a distinction hidden by ordinary packed RAG. In a causal sequence $D_1 D_2 D_3 Q$, later documents acquire hidden states conditioned on earlier documents. Separately encoding D_1, D_2, D_3 does not produce the same state, even if stored keys are later assigned the same rotary positions.

This paper isolates that boundary. Candidate and selection receipts are frozen before comparing text and native execution. The resulting story has three parts: contiguous native transport works; positional rebinding is not the main independent-composition problem; and inexpensive composition repair remains unqualified.

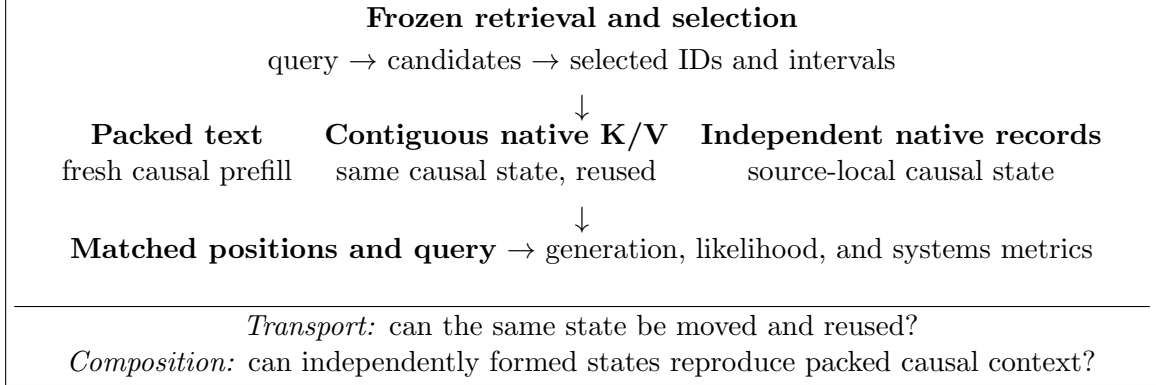


Figure 1: Experimental boundary. Retrieval is computed once. Text and native conditions receive identical selected evidence; only realization and causal history differ.

Contributions.

1. We establish selector-frozen contiguous text/native equivalence in 600 deterministic checks across five replicated natural cohorts.
2. We compare native reuse with both re-prefill and an exact ordinary prefix cache, then measure non-prefix reuse when selected sets change.
3. We introduce a packed/blocked/independent intervention that separates request-time RoPE transport from cross-document contextualization.
4. We qualify host-RoPE rebinding across four precision regimes.
5. We evaluate partial materialization, fixed sparse interaction, parameter-free repair, selective re-encoding, and a bounded learned residual.
6. We publish immutable receipts while keeping retrieval quality orthogonal to native realization.

2 Background and Related Work

RAG combines parametric generation with retrieved evidence [6]. Dense passage retrieval, late interaction, reranking, and multi-hop retrieval improve discovery [3, 4, 12, 13]. Our study does not propose a retrieval leaderboard: a mature lexical, dense, hybrid, or reranked selector can feed the same PRA realization contract.

Serving systems reuse ordinary prefix K/V and manage it in paged storage [5, 2]. CacheBlend recomputes selected positions when fusing cached RAG knowledge [14]; KV-selection and quantization methods reduce active or resident state [16, 7, 11, 8]. PRA instead asks whether selected state can persist as a logically addressed, non-prefix object.

RoPE rotates queries and keys by position [10]. Changing a cached key’s phase can change its effective position, but cannot recreate hidden-state context absent at encoding. Sparse long-context models show that dense interaction is not always necessary [1, 15, 9]; they do not imply that independently contextualized K/V equals a packed causal computation.

Table 1: Principal realization and causal conditions.

Condition	Document computation	Purpose
Packed text (A)	One ordinary causal sequence	Reference RAG behavior
Contiguous native	A’s K/V retained unchanged	Pure transport and reuse
Blocked packed (B)	Same shape/positions; no document-to-document edges	Remove cross-document context
Independent pre-RoPE (C)	Each record encoded alone, rebound to B’s positions	Test transport after causal isolation
Repair	C plus fixed, recomputed, or learned interaction	Measure composition frontier

3 Problem Formulation

Let a frozen selector return ordered intervals $S = (D_1, \dots, D_n)$. *Packed RAG* serializes S and computes

$$H_i^{\text{pack}} = \text{Enc}(D_i \mid D_1, \dots, D_{i-1}).$$

Contiguous native reuse stores the K/V produced by this same packed computation. *Independent native records* instead store

$$H_i^{\text{ind}} = \text{Enc}(D_i).$$

In general, $KV_i^{\text{ind}} \neq KV_i^{\text{pack}}$ because the hidden state projected into K/V has a different causal history. This is a mechanistic distinction tested here, not a universal impossibility theorem.

Each record retains source positions p_{src} . At request time, GLOBAL_PACKED assigns composition positions p_{cmp} . For retained pre-RoPE keys,

$$K_{\ell i}^{\text{request}} = R_{\ell}(p_{\text{cmp}, i})K_{\ell i}^{\text{pre}}.$$

The runtime fingerprints the host frequency tensor, rotary dimensions, scaling policy, and rotation layout; a mismatched contract is rejected.

Figure 1 separates *transport*—whether identical causally formed state can be reused—from *composition*—whether independently formed state can reproduce a packed causal prefill.

4 Experimental Setup

Data and selection. The principal study uses MultiHop-RAG natural questions and a fixed strong reranker under bounded token budgets. Five cohort seeds {11, 23, 37, 71, 101} contain 30 questions each for transport. Candidate, selection, and materialization receipts preserve query IDs, source hashes, token intervals, order, and model revisions.

Models and execution. The powered replicated experiments use pinned Qwen3-1.7B-4bit on Apple MLX. Reduced Qwen3-4B, Qwen3-8B, and Llama-3.1-8B pilots test scale and family boundaries. The base model remains frozen except for the explicitly labeled rank-8 residual experiment.

Metrics and statistics. Quality uses generated token F1, official score, gold-answer mean NLL, output agreement, and first-step JS. Systems metrics keep visible prompt tokens, native tokens, encoding, TTFT, and total latency disjoint. The statistical unit is the seed-cohort mean; 95%

Table 2: Five-seed strong-selector transport summary. Text and native quality are identical within every pair.

Realization/regime	Visible/native	TTFT (ms)	Total (ms)	Pair parity
Selected text, cold	2109/0	1007	1103	–
Native K/V, cold	80/2029	1016	1143	300/300
Selected text, re-prefill	2109/0	1006	1100	–
Native K/V, warm	80/2029	77	203	300/300
Selected text, exact-prefix hit	2109/0	72	165	–

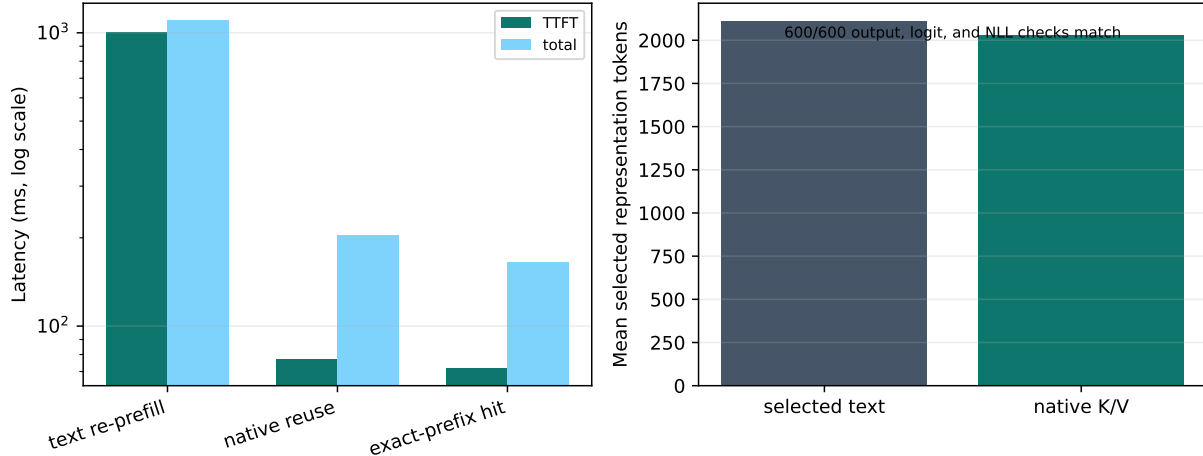


Figure 2: Transport and systems result. Warm native reuse removes evidence re-prefill while preserving all measured outputs, logits, and NLLs. Exact-prefix caching remains faster for an unchanged serialized prefix.

intervals use 10,000 deterministic bootstrap resamples. The 600 transport pairs are deterministic checks, not 600 independent population replicates. Evidence is labeled **Established**, **Qualified boundary**, **Negative**, or **Pilot / uncertain**.

5 Result I: Persistent Native Transport

Across generic and strong selectors, cold and warm execution, all 600 matched text/native pairs preserved receipts, selected intervals, generated output, first-step-logit hashes, token F1, and answer NLL. The maximum NLL difference was zero. This is an **Established** transport result across five natural cohorts.

Relative to text re-prefill, warm native reuse reduces TTFT by 92.3% and total latency by 81.5%. Cold native execution is 0.8% slower in TTFT and 3.7% slower end to end. An exact prefix hit is 6.9% faster in TTFT and 18.9% faster in total latency than warm native reuse. Native transport does not beat a perfect exact-prefix cache.

In 25 four-turn sequences with changing selections, ordinary exact-prefix caching reuses zero evidence tokens. PRA reuses 372.1 of 1325.0 selected K/V entries per turn (28.1%) and reduces measured total time from 716.6 to 557.6 ms (22.2%). This establishes non-prefix addressability, not quality-safe composition: unrepaired rebound reaches .172 F1 versus .217 for fresh packing.

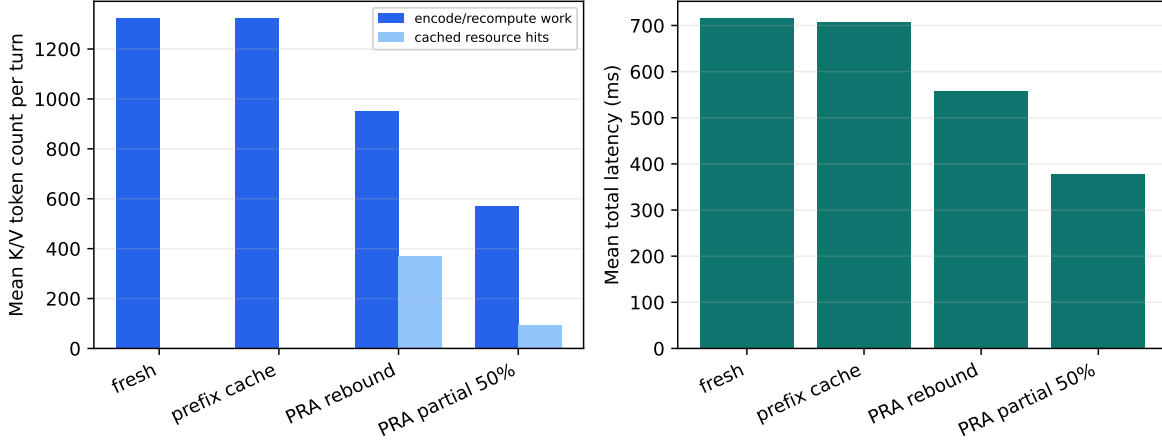


Figure 3: Changing-selection sessions. Exact-prefix reuse is zero, while PRA reuses resource-addressed K/V. Partial disclosure saves more work but does not preserve quality.

6 Result II: Independent Records Lose Causal Context

The central experiment holds token sequence, separators, order, query, and packed positions fixed while changing the document attention graph. A is full packed causal RAG. B prevents document tokens from attending across documents, while query tokens see all documents. C encodes each record independently and applies B’s exact packed positions at request time.

C nearly reproduces B at task level: F1 differs by .0022 and first-step JS is 8.80×10^{-4} . The larger change occurs between A and B, where cross-document causal contextualization is removed. Position is not irrelevant; it is simply not the missing interaction.

A stricter P2 control retains pre-RoPE keys inside B’s single packed tensor and reapplies host RoPE without changing segmentation, projection shape, or attention shape. P2 is exact at FP32, FP16, INT8, and INT4. Separately shaped B/C execution has small early-layer differences that grow through the 4-bit decoder, yielding 67.3% output agreement. Rebinding and numerical shape-path qualification are separate gates; neither reconstructs A’s causal history.

Order controls reinforce this conclusion. With selected identities fixed, canonical, reverse, and random packing change behavior because causal document history changes. Independent records are less order-sensitive because they remove those edges, but lower sensitivity is not a quality win by itself.

Table 3: Five-seed causal decomposition. JS uses A as reference for B and B as reference for C, matching each intervention.

Condition	Token F1	Official	Gold NLL	First-step JS
A: full causal packed	.1187	.467	5.940	—
B: packed, cross-doc blocked	.1103	.400	3.486	.1601 vs A
C: independent, exact offsets	.1081	.393	3.474	.00088 vs B

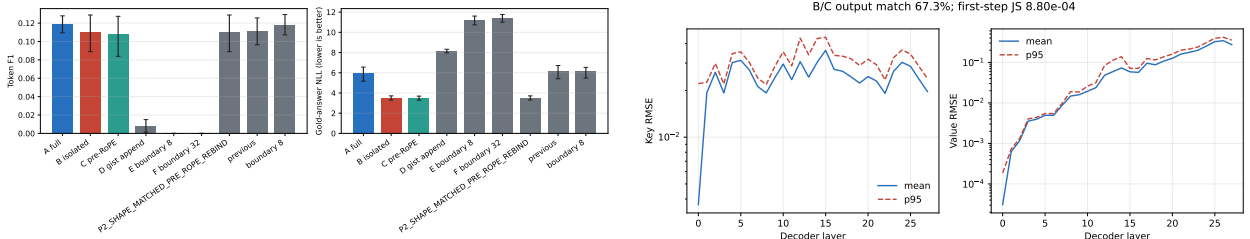


Figure 4: Causal decomposition. Left: removing cross-document edges changes quality; parameter-free repair does not recover it. Right: small B/C numerical differences grow through the quantized decoder.

7 Result III: Composition Repair Frontier

Score-ranked 75% disclosure reduces active K/V from 1320.4 to 990.3 and TTFT from 88.2 to 68.6 ms, but F1 falls from .234 to .181. Smaller budgets are non-monotonic. An equal-budget evidence oracle improves support coverage at 25% (.393 versus .318) but not full-context F1; wrong memory reduces support coverage to .132. This is a **Negative** fixed-policy result.

Appending one mean K/V gist per record, adding all-to-all gist interaction, or copying 8 or 32 boundary tokens uses only 4–256 ephemeral tokens and 16–272 request-specific interactions. Yet mean F1 falls from .108 for independent pre-RoPE records to .007, .000, and .000. Fixed sparse document interaction is also non-monotonic. More interaction is not sufficient; added state must be calibrated to the frozen decoder.

A zero-initialized rank-8 residual (1.38M parameters) is trained on 12 examples from three seeds and evaluated on 30 disjoint examples across five new seeds. It raises F1 from .1352 to .1535 and official score from .433 to .533, while first-step JS falls from .1509 to .0907. However, its paired F1 change of +.0183 has interval $[-.0136, .0558]$, NLL worsens from 2.929 to 3.330, and packed RAG remains at .1986 F1 and .700 official score.

Selective boundary re-encoding is the nonparametric comparator. Re-encoding 16 boundary tokens per record reaches .1525 F1, but official score stays .433, NLL worsens to 5.178, and request transformation costs 191 ms. Neither method passes a joint generation, likelihood, and systems gate.

8 Discussion

Persistent K/V is contextual state, not document content. Contiguous transport succeeds because it preserves state from the same causal computation. Packed documents have directed edges $D_1 \rightarrow D_2 \rightarrow D_3 \rightarrow Q$; independent records meet only at Q . Matching RoPE coordinates does not insert missing arrows.

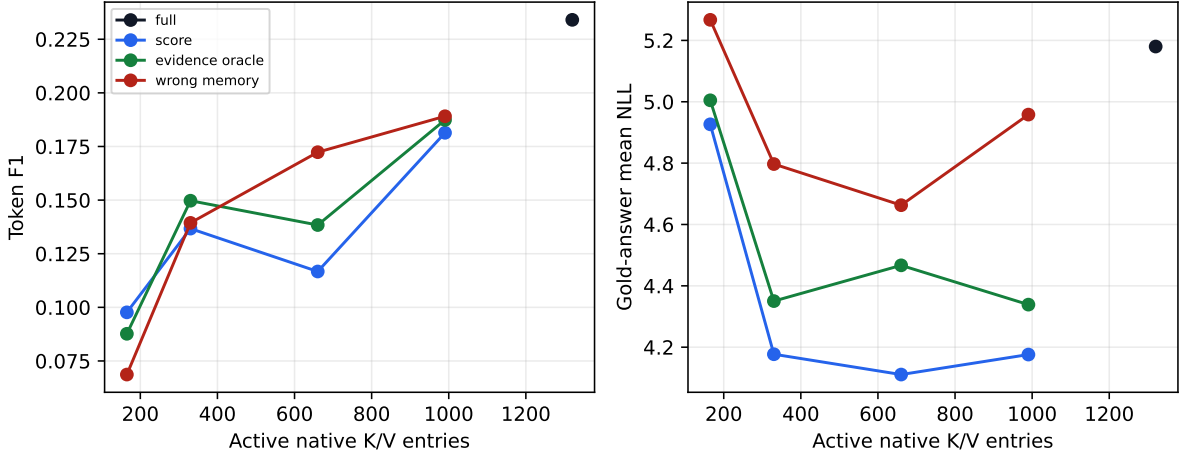


Figure 5: Partial-materialization frontier. F1 and NLL are non-monotonic in active K/V; evidence coverage and generated behavior cannot be inferred from token count alone.

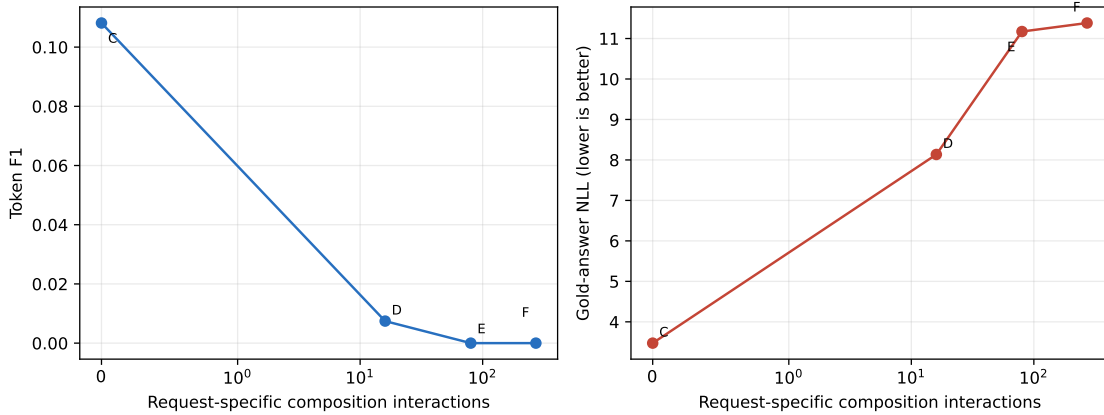


Figure 6: Parameter-free gist and boundary composition. More request-specific interactions sharply degrade F1 and NLL. Arbitrary synthetic K/V is not a safe repair.

Retrieval and realization are orthogonal. PRA need not replace BM25, FAISS, Elasticsearch, Qdrant, a hybrid retriever, or a cross-encoder. Retrieval metrics cannot substitute for transport parity, and transport parity cannot excuse weak selection.

The opportunity begins beyond exact prefixes. An exact prefix hit is the stronger baseline for an unchanged serialized prefix. PRA’s distinct capability is query-addressed reuse when documents recur under changed questions, orders, or combinations. The current independent representation exposes such reuse but does not preserve packed quality.

The evidence motivates a two-part design:

persistent intrinsic state + cheap request-dependent contextualization.

The second term may be learned sparse interaction, selective recomputation, or another bounded adapter. It must pass generation, likelihood, causal, and systems checks together; likelihood proximity alone is insufficient.

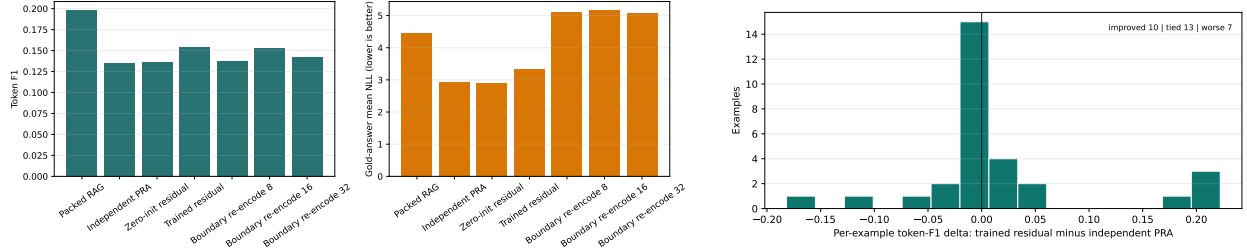


Figure 7: Learned-repair frontier. Aggregate F1 improves while NLL worsens and packed RAG remains stronger (left). Per-example deltas show 10 improvements, 13 ties, and 7 regressions (right); seed-level uncertainty governs the claim.

Table 4: Representative held-out repair frontier.

Condition	F1	Official	NLL	JS	Re-encoded
Packed RAG	.1986	.700	4.467	.0000	0
Independent PRA	.1352	.433	2.929	.1509	0
Trained rank-8 residual	.1535	.533	3.330	.0907	0
Boundary re-encode 16	.1525	.433	5.178	.0876	40
Boundary re-encode 32	.1424	.467	5.077	.0795	80

9 Limitations

The principal composition study uses one 1.7B 4-bit family on bounded natural cohorts. Larger Qwen and Llama experiments are pilots. Separately shaped low-precision execution remains non-identical even though shape-matched rebinding passes. The learned repair has little training data and its five-seed F1 interval crosses zero. Fixed policies do not exhaust all sparse methods. Finally, selector freezing intentionally isolates realization from retrieval; it does not show that PRA retrieval beats strong RAG or prove a universal theorem.

10 Conclusion

Frozen RAG-selected text can be transported into persistent native K/V without changing behavior when its causal context is preserved: all 600 measured pairs match in output, logits, and answer NLL. This gives large warm reuse benefits relative to re-prefill and non-prefix reuse where exact-prefix caching cannot match. Independently encoded records are different. RoPE rebinding restores position but not cross-document context that never occurred. Fixed and parameter-free repairs fail; a small learned residual shows uncertain headroom. Persistent contiguous transport is established in the measured setting. Cheap, quality-preserving composition remains open.

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A Full Retrieval Study

The selector-frozen contract accepts local BM25, hashed dense, FAISS/BGE, reciprocal-rank fusion, BGE and cross-encoder rerankers, Elasticsearch BM25, Qdrant/BGE, and a service hybrid. The service hybrid reaches .852 supporting document Recall@20; Qdrant/BGE reaches .743 and Elasticsearch .735. These qualify candidate generation, not native realization.

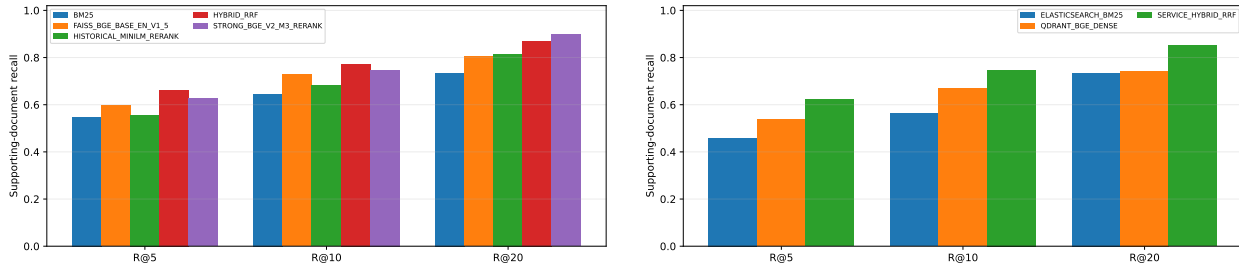


Figure 8: Local (left) and service-backed (right) retrieval ladders.

Service latency is decomposed into embedding, search, and client overhead. Elasticsearch averages 11.5 ms, Qdrant/BGE 23.4 ms including embedding, and service hybrid RRF 53.1 ms on the 50-question cohort. The full report retains backend versions, index revisions, and extended tables.

B Dataset, Cohorts, and Statistics

The dataset revision is fcb9efe65c7730dd4126a42afb6c2c7e45721ebb; the corpus revision is bb98345ef3921312aad05fd117b5a3e39888de2c. Transport uses five independently sampled 30-question cohorts. Repair calibration and evaluation are disjoint. The residual trains on 12 examples associated with seeds 11, 23, and 37 and evaluates on 30 different examples associated with seeds 131, 151, 181, 211, and 223.

Each seed-cohort mean is one statistical unit. Bootstrap intervals resample the five seed means 10,000 times with a fixed random seed. Raw or compressed per-example rows remain committed beside each manifest.

C Full Transport and Cache Controls

The 600 pairs comprise 300 generic-selector and 300 strong-reranker comparisons. The reduced 4B/8B/family study adds 120 exact pairs. At 4B, Qwen-8B, and Llama-8B, warm native TTFT is

6.5%, 5.9%, and 6.0% of text re-prefill, but 7.2%, 4.6%, and 5.0% slower than exact-prefix reuse. These scale rows are **Pilot** / **uncertain** because each uses one 20-question cohort.

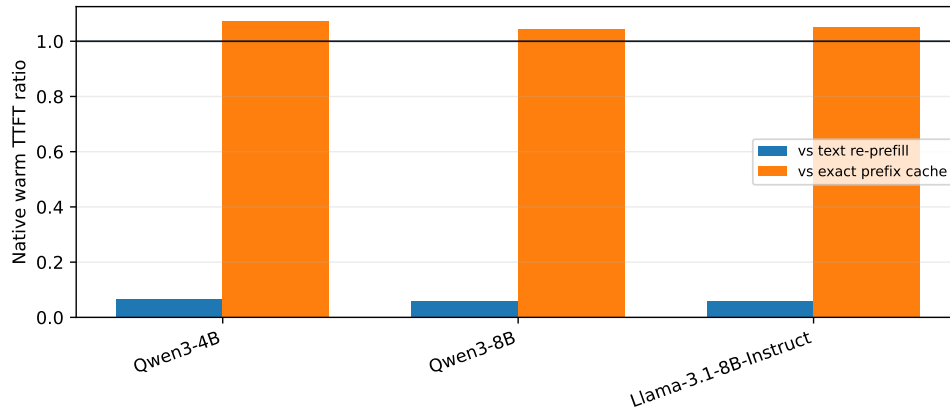


Figure 9: Cross-scale transport pilot. Native reuse is far cheaper than re-prefill but slightly slower than a perfect prefix hit.

D Positional Transport and Precision

The runtime captures projected keys before host RoPE and keeps values unchanged. P2 changes only storage/rebinding and is exact at FP32, FP16, INT8, and INT4. Separately shaped record projection and attention calls introduce precision-dependent drift that compounds by layer.

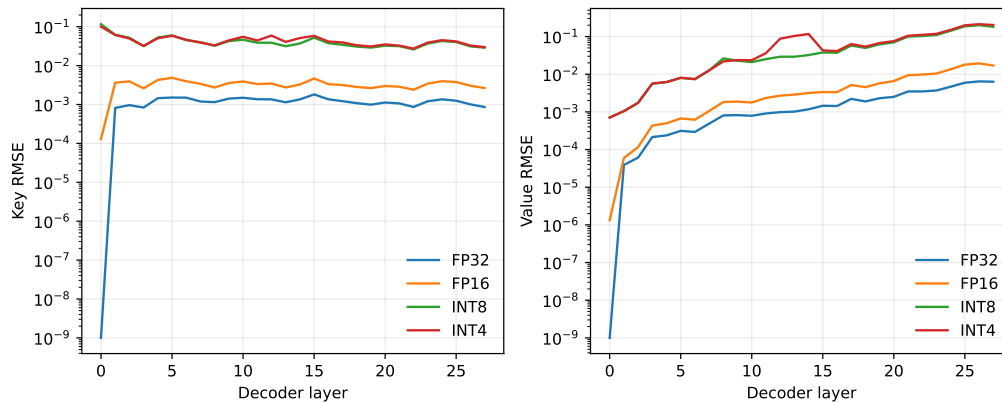


Figure 10: Layerwise B/C precision diagnostic. Shape-matched rebinding is exact; separate record-wise execution introduces drift.

The initial Llama pilot incorrectly assumed one RoPE base and produced JS .526. The corrected path reads the host layer’s piecewise frequency tensor. Only corrected values appear here.

E Independent-Record Diagnostics

Independent records use the canonical URI-addressed `ContextRecord` and resolver path. Explicit and rooted references agree exactly, excluding parser or root resolution as the measured loss. Stronger

selectors improve evidence coverage, but independent realization remains below packed RAG in replicated 1.7B and 4B studies.

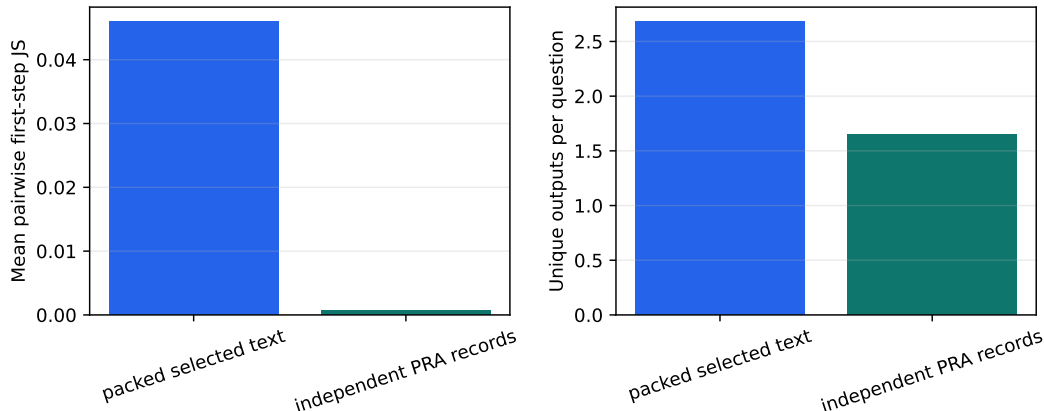


Figure 11: Packed and independent-record order sensitivity. Reduced sensitivity from removing cross-document edges is not itself a quality gain.

Canonical, reverse, and seeded-random orders freeze identities and content. Packed results vary because causal history varies; independent records vary less because documents meet only at the query.

F Partial Materialization and Sparse Interaction

Score, evidence-oracle, and wrong-memory policies are evaluated at 12.5%, 25%, 50%, and 75%. None gives a monotonic F1/NLL frontier. Fixed masks include previous-document, top-ranked-to-all, and 8/16/32/64-token boundaries. The eight-token window uses 99.97% fewer cross-document edges than full packing and approaches its mean F1, but fails official-score and NLL gates.

A diagnostic 25% repair raises changing-selection output recovery to 44% but requires 1284.1 newly encoded or recomputed states, only 3.1% less than fresh packing. Partial 50% disclosure cuts new work by 57.1% and measured time by 47.2%, but reaches only .150 F1 and 6% parity.

G Parameter-Free Repair Attempts

The family includes mean K/V gists, all-to-all gist mixing, first/last boundary copies, and request-local 8- or 32-token windows. It never overwrites persistent records and uses one host attention normalization over persistent, ephemeral, and query state. Every variant fails the five-seed gate and reduced Qwen/Llama pilots. This rules out a tempting shortcut: arbitrary synthetic K/V is not automatically meaningful to a frozen decoder.

H Learned Residual Repair

The rank-8 adapter is zero-initialized and optimized for 100 AdamW steps at 2×10^{-4} with K/V distillation, response distillation, and 0.1 task-loss weight. The paired F1 deltas versus independent PRA are +.0633 for packed RAG ([.0264,.1170]), +.0183 for the trained residual ([-.0136,.0558]),

+0.0173 for 16-token re-encoding $([-.0230, .0474])$, and +0.0072 for 32-token re-encoding $([-.0184, .0328])$. With five seeds, the trained effect remains **Pilot / uncertain**.

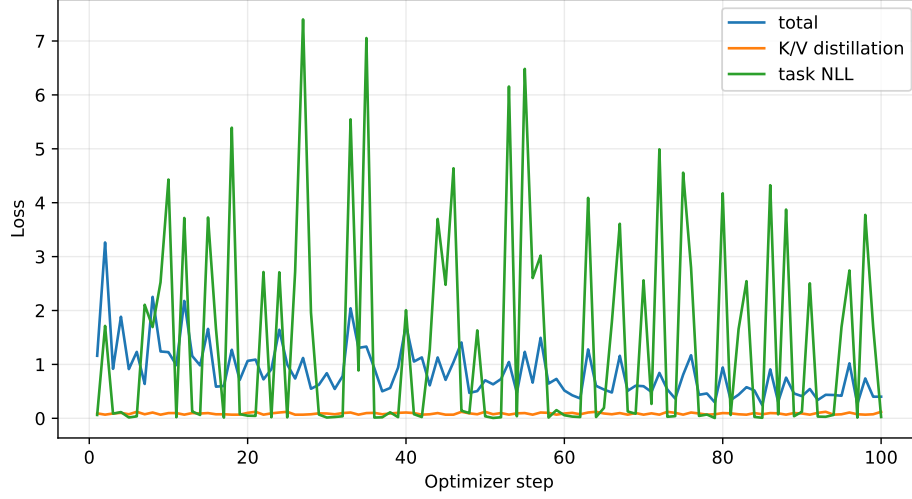


Figure 12: Residual calibration history. Step 100 is selected; evaluation data and seeds are disjoint from calibration.

I Cross-Model Pilots

On five questions in canonical and reverse order, 50% later-resource prefix repair recovers 8/10 Qwen3-4B, 5/10 Qwen3-8B, and 7/10 Llama-3.1-8B outputs. Packed rebinding alone recovers 2/10, 0/10, and 3/10; full repair reaches 10/10 for all. This does not establish a monotonic scale law.

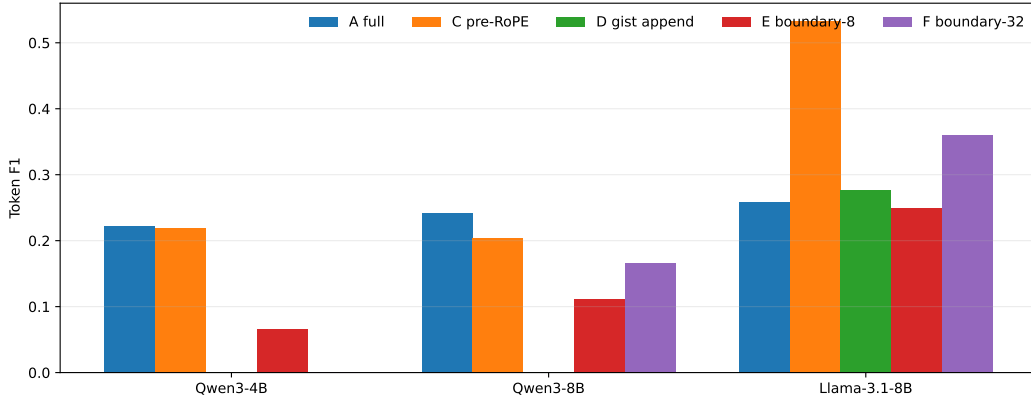


Figure 13: Reduced cross-model composition pilot.

J Systems, Artifacts, and Evidence Ledger

Runs persist dataset/corpus revisions, query identity, candidate and selection receipts, scores, record IDs, spans, order, positions, precision, generation settings, backend versions, hardware, and commit

ID. Native records use immutable URIs and source hashes. Overlap is deduplicated only within one source domain.

The compact entry point is `docs/papers/shared/results/paper3_2_rag/publication/publication_summary.json`. Figures are regenerated by `experiments/paper3_2_rag/build_publication_artifacts.py`. The exhaustive experimental sequence, configurations, and tables remain in `paper_3_2_full.pdf`.

Table 5: Evidence hierarchy after reorganization.

Status	Result
Established	600/600 contiguous checks; shape-matched host-RoPE rebinding; exact-prefix and changing-selection cache controls
Qualified boundary	Independent records track blocked packed context; separately shaped INT4 is approximate; sparse interaction is non-monotonic
Negative	Fixed partial materialization; parameter-free gist append; post-hoc boundary copies
Pilot / uncertain	4B/8B and Llama composition; rank-8 residual with interval crossing zero